Braille Tech: IoT-Enabled Assistive Device

Braille Device for the Visually Impaired

GROUP MEMBERS:

VILLEGAS, JEAN KATHLEEN

TEMONES, ANGELICA D.

LOSORATA, JERAMIE ANNE Q.

Project Description

Braille Tech is an innovative assistive device designed to enhance accessibility for visually impaired individuals by translating digital text into tactile Braille in real time. Leveraging the power of the Internet of Things (IoT), this device seamlessly integrates modern communication protocols like MQTT with robust hardware components, including microcontrollers and servo motors. The result is a compact, user-friendly, and cost-effective solution that bridges the gap between digital information and physical accessibility.

The device operates by receiving text input through a connected IoT network, processing it into Braille characters, and rendering these characters onto a tactile interface using servo motors. Users can touch the Braille interface to read the text, ensuring a smooth and empowering reading experience.

This project not only demonstrates the potential of IoT in assistive technology but also provides a scalable foundation for further enhancements, such as text-to-speech capabilities or integration with smart devices like smartphones and home assistants.

Key Features

1. **Seamless Text Input**: Supports text entry via Wi-Fi (HTTP server) or MQTT protocol for efficient communication.
2. **Real-Time Conversion**: Maps input text to Braille using an onboard lookup table for instant processing.
3. **Dynamic Braille Display**: Uses servo-driven tactile horns in a 2x3 grid to represent Braille characters sequentially.
4. **User-Friendly Interaction**: Includes physical buttons and web-based commands for navigation, such as displaying the next character or resetting the system.
5. **Lightweight Communication**: MQTT ensures low-latency performance, even in limited bandwidth environments.

Functionality

Text Input and Communication

The device uses the ESP32 Devkit V1 board as its central processing unit, leveraging its built-in Wi-Fi and Bluetooth capabilities for receiving text data. Text can be sent to the device through various methods:

- **Wi-Fi:** A web-based interface allows users or caretakers to input text via an HTTP server.
- **MQTT Protocol:** The system uses MQTT for lightweight, low-latency communication to fetch or transmit data from a centralized server. This ensures the system remains responsive even in environments with limited bandwidth.

Text-to-Braille Conversion

The received text data is processed in real-time by the ESP32. Each character is mapped to its corresponding Braille representation using a preloaded lookup table stored in the ESP32's memory.

Braille Tactile Display

The tactile Braille display consists of servo motors that control small, raised horns to physically represent Braille patterns.

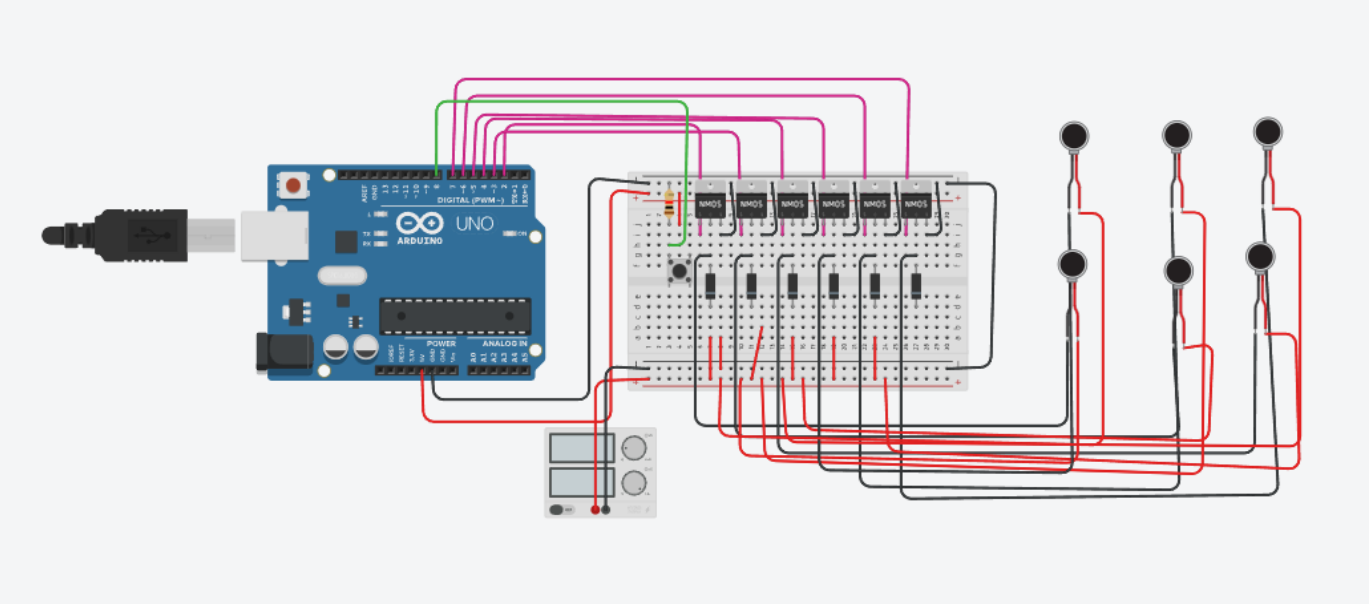
- Each Braille character is represented by six servo-controlled horns arranged in a 2x3 grid.
- The servos raise or lower its tactile horns based on the Braille pattern for the given character.
- This dynamic system allows the device to render a wide range of characters sequentially (letters A-Z).

User Interaction

The device incorporates physical buttons and web-based commands for intuitive navigation:

1. **Physical Button:**
    - **Primary Button:** Pressing the button initializes the device to start and displays the first Braille character, "A".
2. **Web Interface Commands:**
    - **"Next Letter":** Moves to the subsequent character in the text and updates the Braille display accordingly.
    - **"Reset":** Resets the system to the starting point (displaying none) and clears any previously loaded text.

Arduino Simulation



Materials Used

1. **USB Cable**
    - **Function:** Used to program the Arduino UNO and ESP32, and supply power during testing.
2. **Micro Servo SG90 9g (6 pcs)**
    - **Function**: Actuates the Braille pins to create tactile feedback for each Braille character.
    - **Why?** Lightweight and precise, ideal for simulating Braille patterns. Each servo operates a pin or a set of pins for the Braille display.
3. **Breadboard**
    - **Function:** Provides a temporary platform for assembling and testing the circuit connections.
4. **Jumper Wires (Female to Male)**
    - **Function:** Used to connect components on the breadboard and to the microcontrollers.
5. **Push Button**
    - **Function:** Acts as a manual input for controlling the device or resetting its operations.
6. **ESP32 DevKit V1 Board – Tag 370**
    - **Function:** Provides wireless connectivity (Wi-Fi) for IoT functionality, enabling the device to receive text data over a network.
    - **Why?** The ESP32 is a versatile microcontroller with built-in Wi-Fi and Bluetooth, perfect for IoT applications